# Kunal Shukla

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## Education

**SRM Institute of Science and Technology, Kattankulathur** 2025 – 2029 (expected)
*B.Tech, Computer Science and Engineering (Data Science)* CGPA: 8.14 / 10

## Relevant Coursework

- Data Structures & Algorithms
- Fundamentals of Data Science
- Calculus & Linear Algebra
- Transforms & Boundary Value Problems
- Advanced Calculus & Complex Analysis
- Object-Oriented Design & Programming
- Operating Systems
- Advanced OOP (Java)
- Programming for Problem Solving

## Projects

**m2-infer** | *LLM inference engine from scratch: Python, MLX, Metal, GGUF* **GitHub | Sep 2026**

- Built a **Qwen2 inference engine from scratch**: GGUF reader, **bit-exact dequantisers**, BPE tokenizer and **KV cache**, matching Hugging Face Transformers' logits to $5 \times 10^{-4}$. **4-bit decode** runs **66 tok/s** on Qwen2.5-1.5B on a fanless M2 (llama.cpp 57, MLX-LM 85; same prompt and context).
- Measured the **hardware ceiling** (90.8 GB/s, 89% of the memory bus) and fit a **per-token cost model** that predicts a left-out model's speed within **6–8% median error** (10 models each for llama.cpp and Ollama).
- Wrote **Metal GPU kernels** and serving: a **paged-attention** kernel **2**× faster than copy-out (0.19 vs 0.39 ms), **speculative decoding** up to **1.92**× on 7B, and an OpenAI-compatible server whose **continuous batching** raised requests meeting the latency SLO at 1 req/s from 7% to **87%**.

**AcadKit** | *Full-stack + AI: TypeScript, React, PostgreSQL, Go, local VLMs* **GitHub | Live | Feb 2026 – Present**

- Shipped **v1.0.0** of a phone + Mac app for SRM students (attendance, marks, 5-day Day-Order timetable): a **PWA** on Postgres with **row-level security** on every table, 62 migrations and **CI/CD** with approval-gated deploys.
- A **local vision-language pipeline** (Qwen3-VL) reads 110 scanned past papers: it finds **76.2%** of 2,409 labelled questions (84.6% in end-sems) and reads marks **87.8%** right; **embeddings** cross-link 7,213 notes, slides and questions.
- Grade solver checked against **brute force** on 6,000 mark plans and 14,760 attendance cases; **1,531 unit tests**, **end-to-end tests** on Chromium, iPhone and Mac, Lighthouse 100.
- Built **jobq** (Go), the job queue that delivered its push reminders: `SKIP LOCKED` claims with **leases and retries** ran **5,071 jobs/s** on 256 workers (99% of ideal), and my own **append-only log storage engine** reached **27**× throughput with **group commit**, crash-tested with hundreds of SIGKILLs.

**How Vision-Language Models See Depth** | *AI research: PyTorch, three.js* **In progress | 2026**

- Built a seeded **3D renderer** and **adaptive Bayesian staircases** (Psi method) to measure how VLMs use five depth cues; validated on **simulated observers** (90–98% CI coverage) and ran **9 models**, local and hosted.
- Found Qwen2.5-VL-3B reads **relative size backwards** (23 of 118 correct, $p \approx 7 \times 10^{-12}$), unchanged at fp16, so **not a quantisation artefact**; Qwen2.5-VL-7B sees all five cues but uses **only height** to judge depth.

## Technical Skills

**Languages & Tools:** Python, C, Java, SQL, TypeScript, Go, Bash, Git, Linux, Playwright, Vitest
**ML & Data Science:** PyTorch, Hugging Face Transformers, pandas, NumPy, SciPy, Matplotlib, Seaborn
**LLM Systems:** MLX, Metal, GGUF, quantisation, KV cache, paged attention, speculative decoding, Ollama
**Backend & DevOps:** PostgreSQL, Supabase, REST APIs, Node.js, React, GitHub Actions, Prometheus, k6

## Open Source & Certifications

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| **Hono:** | Found bugs by **differential fuzzing**: a **wildcard-routing bug** whose fix was merged into v5 with me as **co-author** (#5556), a query-parsing bug fixed in v4.13.14 (#5580), and an open middleware-skip bug (#5544) |
| **Apache Spark:** | Diagnosed and reported a **pandas 3.1.0** import break in `pyspark.pandas` (removed pandas internals) with a minimal repro (#59824) |
| **Security:** | Reported security issues across the **Koa ecosystem** (@koa/router authorization skip, signed-cookie, CSS injection) and Hono by **private GitHub Security Advisories** |
| **Oracle, 2026:** | OCI Architect Associate (96%); AI Foundations (95%); Agentic AI Foundations (98%); AI Database Foundations (100%); OCI Foundations (80%) |
| **Mercor, 2026:** | Introduction to Agents — AI agent evaluation (rigor, calibration, clarity) |
| **DataCamp, 2026:** | GitHub Foundations (9-hour course) |